

Certificate generation Screenshot:

```
Loading existing Certificate Authority...
CA: O=Grand Marina Hotel,OU=Water Systems Security,CN=Grand Marina Root CA

Server certificate exists: certs/server.pem

--- Generating certificate for HYDROLOGIC-MainBuilding-001 ---
Generating private key...
Creating certificate...
Certificate saved: certs/device-001.pem
Private key saved: certs/device-001-key.pem

--- Generating certificate for HYDROLOGIC-PoolSpa-002 ---
Generating private key...
Creating certificate...
Certificate saved: certs/device-002.pem
Private key saved: certs/device-002-key.pem

--- Generating certificate for HYDROLOGIC-Kitchen-003 ---
Generating private key...
Creating certificate...
Certificate saved: certs/device-003.pem
Private key saved: certs/device-003-key.pem

=====
All certificates generated successfully!
=====

Certificate Summary:
-----
Certificate Authority:
Certificate: certs/ca.pem
Private key: certs/ca-key.pem

Server (Mosquitto broker):
Certificate: certs/server.pem
Private key: certs/server-key.pem

Client Devices:
HYDROLOGIC-MainBuilding-001:
Certificate: certs/device-001.pem
Private key: certs/device-001-key.pem
HYDROLOGIC-PoolSpa-002:
Certificate: certs/device-002.pem
Private key: certs/device-002-key.pem
HYDROLOGIC-Kitchen-003:
Certificate: certs/device-003.pem
Private key: certs/device-003-key.pem

Next steps:
1. Start Mosquitto: mosquitto -c mosquitto_tls.conf -v
2. Run subscriber: python subscriber_mtls.py
3. Run publisher: python publisher_mtls.py
```

Broker Startup Screenshot:

```
3. Run publisher: python publisher_mtls.py
kaylacolletti@Mac hydroficient-project % mosquitto -c mosquitto_tls.conf -v
1775710629: Info: running mosquitto as user: kaylacolletti.
1775710629: mosquitto version 2.1.2 starting
1775710629: Config loaded from mosquitto_tls.conf.
1775710629: Bridge support available.
1775710629: Persistence support available.
1775710629: TLS support available.
1775710629: TLS-PSK support available.
1775710629: Websockets support available.
1775710629: Opening ipv6 listen socket on port 8883.
1775710629: Opening ipv4 listen socket on port 8883.
1775710629: mosquitto version 2.1.2 running
1775711123: New connection from ::1:63784 on port 8883.
1775711123: New client connected from ::1:63784 as GrandMarina-Dashboard (p4, c1, k60, u'HYDROLOGIC-MainBuilding-001').
1775711123: No will message specified.
1775711123: Client GrandMarina-Dashboard negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1775711123: Sending CONNACK to GrandMarina-Dashboard (0, 0)
1775711123: Received SUBSCRIBE from GrandMarina-Dashboard
1775711123:   hydroficient/grandmarina/# (QoS 1)
1775711123: GrandMarina-Dashboard > hydroficient/grandmarina/#
1775711123: Sending SUBACK to GrandMarina-Dashboard
1775711139: New connection from ::1:63785 on port 8883.
1775711139: New client connected from ::1:63785 as HYDROLOGIC-Device-001 (p4, c1, k60, u'HYDROLOGIC-MainBuilding-001').
1775711139: No will message specified.
1775711139: Client HYDROLOGIC-Device-001 negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1775711139: Sending CONNACK to HYDROLOGIC-Device-001 (0, 0)
1775711140: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m1, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711140: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m1, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711140: Sending PUBACK to HYDROLOGIC-Device-001 (m1, rc0)
1775711140: Received PUBACK from GrandMarina-Dashboard (Mid: 1, RC:0)
1775711145: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m2, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711145: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m2, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711145: Sending PUBACK to HYDROLOGIC-Device-001 (m2, rc0)
1775711145: Received PUBACK from GrandMarina-Dashboard (Mid: 2, RC:0)
1775711150: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m3, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711150: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m3, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1775711150: Sending PUBACK to HYDROLOGIC-Device-001 (m3, rc0)
1775711150: Received PUBACK from GrandMarina-Dashboard (Mid: 3, RC:0)
1775711155: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m4, 'hydroficient/grandmarina/device-001/sensors', ... (282 bytes))
1775711155: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m4, 'hydroficient/grandmarina/device-001/sensors', ... (282 bytes))
1775711155: Sending PUBACK to HYDROLOGIC-Device-001 (m4, rc0)
1775711155: Received PUBACK from GrandMarina-Dashboard (Mid: 4, RC:0)
█
```

Successful connection:

Publisher with mTLS connecting successfully-

```
cat /dev/urandom | tr -dc 'a-z0-9' | fold -n 64 | xargs sha1sum
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/subscriber_mtls.py:100: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[TLS] mTLS configured with client certificate

[CONNECTING] localhost:8883...

[LISTENING] Waiting for messages (Ctrl+C to stop)...

[SUCCESS] Connected to broker as GrandMarina-Dashboard
[INFO] Subscribing to: hydroficient/grandmarina/#
[SUBSCRIBED] QoS granted: (1,)

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 01:05:40
-----
Device: HYDROLOGIC-Device-001
Flow Rate: 52.28 LPM
Pressure (Up): 59.88 PSI
Pressure (Down): 54.49 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 01:05:45
-----
Device: HYDROLOGIC-Device-001
Flow Rate: 51.73 LPM
Pressure (Up): 59.87 PSI
Pressure (Down): 55.81 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 01:05:50
-----
Device: HYDROLOGIC-Device-001
Flow Rate: 48.06 LPM
Pressure (Up): 54.44 PSI
Pressure (Down): 54.44 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 01:05:55
-----
Device: HYDROLOGIC-Device-001
Flow Rate: 47.8 LPM
Pressure (Up): 59.69 PSI
Pressure (Down): 54.68 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 01:06:00
-----
Device: HYDROLOGIC-Device-001
Flow Rate: 47.19 LPM
Pressure (Up): 58.54 PSI
Pressure (Down): 57.86 PSI
Status: operational
█
```

Subscriber Receiving Data-

```
cat /dev/urandom | tr -dc 'a-z0-9' | fold -n 64 | xargs sha1sum
kaylacolletti@Mac hydroficient-project % python3 publisher_mtls.py
=====
HYDROLOGIC Sensor Publisher (mTLS)
=====
Device ID: 001
Topic: hydroficient/grandmarina/device-001/sensors
Certificate: certs/device-001.pem
=====
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/publisher_mtls.py:108: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[TLS] mTLS configured with client certificate

[CONNECTING] localhost:8883...
[SUCCESS] Connected to broker as HYDROLOGIC-Device-001
[INFO] Certificate identity verified by broker

[PUBLISHING] Sending sensor readings (Ctrl+C to stop)...

[1] Published: 52.28 LPM
[PUBLISHED] Message ID: 1
[2] Published: 51.73 LPM
[PUBLISHED] Message ID: 2
[3] Published: 48.06 LPM
[PUBLISHED] Message ID: 3
[4] Published: 47.8 LPM
[PUBLISHED] Message ID: 4
[5] Published: 47.19 LPM
[PUBLISHED] Message ID: 5
[6] Published: 47.08 LPM
[PUBLISHED] Message ID: 6
█
```

Rejection Screenshot:

```
kaylacolletti@Mac hydroficient-project % python3 test_no_cert.py.txt
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/test_no_cert.py.txt:11: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="rogue-device", **MQTT_CLIENT_ARGS)
ERROR: Connection should have been rejected!
kaylacolletti@Mac hydroficient-project % █
```

Certs issued by the same CA: CN=Grand Marina Root CA, OU=Water Systems Security, O=Grand Marina Hotel

CA most common: Grand Marina Root CA